

The file works by always listening for a key press, and then if a key press occurs it runs the `draw(screen)` function which updates the display with the new text. The text is edited in the functions that come later (`main()`).

1. The draw function:

The display uses substrings, going up until the cursor's intended index position, and then continuing the text.

```
def draw(screen):
    screen.clear()

    # DRAW DISPLAY

    display = text[:cursor] + "|" + text[cursor:]

    # -----

    for row, line in enumerate(display.split("\n")):
        screen.addstr(row, 0, line)

    screen.addstr(
        len(display.split("\n")) + 1,
        0,
        "← → Move   Type Insert   Backspace Delete   Enter New Line   Esc Quit"
    )

    screen.refresh()
```

2. The main function:

The main function listens for key presses. When a user presses the left arrow, the cursor idx subtracts one (moving left), the same process happens for the right arrow but the cursor idx is added by one.

The backspace function will do the same thing as the left arrow, but will ignore the character that was just to the left of the cursor when updating the text to simulate a deletion.

New line and insert work similarly, adding another character before adding the cursor.

```

def main(screen):
    global text, cursor

    while True:
        draw(screen)

        key = screen.getch()

        if key == 27:
            break

        # LEFT ARROW

        elif key == curses.KEY_LEFT:

            if cursor > 0:
                cursor -= 1

            # -----

            # RIGHT ARROW

            elif key == curses.KEY_RIGHT:

                if cursor < len(text):
                    cursor += 1

                # -----

                # BACK SPACE

            elif key in (8, 127, curses.KEY_BACKSPACE):

                if cursor > 0:
                    text = text[:cursor-1] + text[cursor:]
                    cursor -= 1

                # -----

                # NEW LINE

            elif key == 10:

                text = text[:cursor] + "\n" + text[cursor:]
                cursor += 1

                # INSERT CHARACTER

            elif 32 <= key <= 126:

                text = text[:cursor] + chr(key) + text[cursor:]
                cursor += 1

```